COMPUTER NETWORKING LAB

Cisco Packet Tracer Experiments — Shortened Steps

Experiments 10, 11, 12, 13, 14 and 16

Name: _____

Register No.: _____

Class / Section: _____

Date: _____

Note: After each experiment, take a screenshot (Win + Shift + S) of the listed output and paste it in the record.

Experiment 10: Network Topologies (Bus, Star, Mesh, Ring)

Aim: To draw Bus, Star, Mesh and Ring topologies in Cisco Packet Tracer.

Devices: PCs, hubs, 2960 switches, straight-through and crossover cables.

A. Star Topology

1. Place 1 switch and 4 PCs.
2. Connect each PC FastEthernet0 to the switch using straight-through cable.
3. Set PC0–PC3 IPs to 192.168.1.1–192.168.1.4; mask 255.255.255.0.
4. Wait for green links and ping PC3 from PC0.

B. Bus Topology

1. Place 3 hubs in a line and connect 1 PC to each hub.
2. Connect Hub0–Hub1–Hub2 using crossover cables (backbone).
3. Connect each PC to its hub using straight-through cable.
4. Set IPs 192.168.2.1–192.168.2.3 and ping across the bus.

C. Ring Topology

1. Place 4 switches and 4 PCs (one PC per switch).
2. Connect Switch0 → Switch1 → Switch2 → Switch3 → Switch0 using crossover cables.
3. Connect each PC to its switch using straight-through cable.
4. Set IPs 192.168.3.1–192.168.3.4 and ping.

D. Mesh Topology

1. Place 4 switches and 4 PCs.
2. Connect every switch to every other switch using crossover cables: S0-S1, S0-S2, S0-S3, S1-S2, S1-S3, S2-S3 (6 links).
3. Connect one PC to each switch and set IPs 192.168.4.1–192.168.4.4.
4. Ping to verify connectivity.

Expected output: Successful ping, e.g. 4 sent, 4 received, 0% loss.

Output images: All four topologies with green links; one successful ping window.

Result: The four network topologies were drawn and verified.

Experiment 11: HTTP Traffic Analysis and Cookie Examination

Aim: To analyse HTTP traffic and examine HTTP packet details.

Devices: 1 Server-PT, 1 PC, 1 switch, straight-through cables.

Procedure

1. Connect the server and PC to the switch using straight-through cables.
2. Server: Config → FastEthernet0; IP 192.168.1.10, mask 255.255.255.0.
3. PC: Desktop → IP Configuration; IP 192.168.1.1, mask 255.255.255.0.
4. Server: Config → Services → HTTP; turn HTTP ON (edit index.html if needed).
5. Enter Simulation mode → Edit Filters; select only HTTP and TCP.
6. PC: Desktop → Web Browser; open http://192.168.1.10.
7. Click Capture/Forward until the page loads.
8. Select the Event List envelope → Inbound/Outbound PDU Details; inspect SYN, SYN-ACK, ACK, HTTP GET and HTTP 200 OK.
9. Record source/destination IPs, port 80, HTTP method and headers. Cookie details are inspected in PDU headers; Packet Tracer's browser does not show a real cookie header.

Expected output:

```
PC > Server       TCP    SYN
Server > PC       TCP    SYN, ACK
PC > Server       TCP    ACK
PC > Server       HTTP   GET /
Server > PC       HTTP   200 OK
```

Output images: Simulation Event List; HTTP PDU Details; browser showing server page.

Result: HTTP traffic and request/response packets were examined.

Experiment 12: Configuring DHCP and Static IP in a LAN

Aim: To configure DHCP and static IP addressing in a LAN.

Devices: 1 Server-PT, 1 switch, 3 PCs, straight-through cables.

Procedure

1. Connect the server and all 3 PCs to the switch.
2. Server: Config → FastEthernet0; static IP 192.168.1.1, mask 255.255.255.0.
3. Server: Config → Services → DHCP; turn ON and set Pool Name LAN_POOL, gateway 192.168.1.1, DNS 192.168.1.1, start IP 192.168.1.10, mask 255.255.255.0, Max Users 50; click Save.
4. PC0 and PC1: Desktop → IP Configuration → select DHCP; they should receive addresses such as 192.168.1.10 and .11.
5. PC2: Desktop → IP Configuration → Static; IP 192.168.1.100, mask 255.255.255.0, gateway 192.168.1.1.
6. Verify with ipconfig, then ping 192.168.1.1 and 192.168.1.100.

Expected output:

```
PC0> ipconfig
IP Address......: 192.168.1.10
Subnet Mask.....: 255.255.255.0
Default Gateway.: 192.168.1.1

PC0> ping 192.168.1.100
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
```

Output images: Server DHCP page; DHCP PC IP Configuration; static PC IP Configuration; successful ping.

Result: DHCP and static IP addressing were configured and verified.

Experiment 13: Configuring 2 VLANs

Aim: To configure two VLANs using Cisco Packet Tracer.

Devices: 1 2960 switch, 4 PCs, straight-through cables.

Procedure

1. Connect PC0–PC3 to switch ports Fa0/1–Fa0/4 respectively.

2. Set IPs: PC0 192.168.10.1, PC1 192.168.10.2, PC2 192.168.20.1, PC3 192.168.20.2; mask 255.255.255.0.

3. Open the switch CLI and enter the commands below.

4. Ping PC0 to PC1 (same VLAN), then PC0 to PC2 (different VLAN).

Switch commands

```
enable
configure terminal
vlan 10
name SALES
exit
vlan 20
name HR
exit
interface range fa0/1-2
switchport mode access
switchport access vlan 10
exit
interface range fa0/3-4
switchport mode access
switchport access vlan 20
end
show vlan brief
```

Expected output:

```
VLAN Name    Status   Ports
10   SALES   active   Fa0/1, Fa0/2
20   HR      active   Fa0/3, Fa0/4

PC0> ping 192.168.10.2   Reply (success)
PC0> ping 192.168.20.1   Request timed out (fails)
```

Output images: Topology; show vlan brief; successful same-VLAN ping and failed cross-VLAN ping.

Result: Two VLANs were configured; same-VLAN devices communicate, but different VLANs cannot communicate without routing.

Experiment 14: Point-to-Point and Multipoint Configuration

Aim: To demonstrate point-to-point and multipoint connections.

Devices: PCs, 1 hub, crossover and straight-through cables.

A. Point-to-Point

1. Place 2 PCs and connect them directly with a crossover cable.

2. Set PC0 to 192.168.1.1 and PC1 to 192.168.1.2; mask 255.255.255.0.

3. Ping PC1 from PC0.

B. Multipoint

1. Place 1 hub and 4 PCs; connect each PC to the hub using straight-through cable.

2. Set IPs 192.168.2.1–192.168.2.4; mask 255.255.255.0.

3. Ping from PC0 to the other PCs; in Simulation mode, observe the hub sending data through all ports.

Expected output:

```
PC0> ping 192.168.2.3
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Output images: Point-to-point topology with successful ping; multipoint topology with successful ping.

Result: Point-to-point and multipoint connections were demonstrated.

Experiment 16: Network Transmission Modes

Aim: To demonstrate simplex, half-duplex and full-duplex modes.

Devices: PCs, 1 hub, 1 switch, straight-through and crossover cables.

A. Simplex (one direction)

1. Connect PC0 and PC1 with crossover cable; set IPs 192.168.1.1 and 192.168.1.2.
2. Simulation → Add Complex PDU; set Source PC0, Protocol UDP, Destination 192.168.1.2, Sequence One Shot; click Send.
3. Observe that the packet travels PC0 → PC1 only, with no reply.

B. Half-Duplex

1. Connect 3 PCs to a hub with straight-through cables; set IPs 192.168.2.1–192.168.2.3.
2. For each PC, Config → FastEthernet0 → set Half Duplex.
3. In Simulation mode, ping PC0 → PC1 and PC2 → PC1 at the same time.
4. Observe that only one device transmits at a time; simultaneous transmissions may cause a collision (red cross).

C. Full-Duplex

1. Connect 3 PCs to a switch with straight-through cables; set IPs 192.168.3.1–192.168.3.3.
2. For each PC, Config → FastEthernet0 → set Full Duplex.
3. In Simulation mode, send pings PC0 → PC1 and PC1 → PC0 together.
4. Observe packets moving in both directions simultaneously without collision.

Expected output: Simplex: one-way; half-duplex: one direction at a time/collision possible; full-duplex: both directions simultaneously/no collision.

Output images: Simplex one-way PDU; half-duplex hub topology (collision if shown); full-duplex switch topology; FastEthernet0 duplex setting.

Result: Simplex, half-duplex and full-duplex transmission modes were demonstrated.